

Applied Financial Risk Management

Session 8

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On the first approach

- If returns are assumed to be normal, then the (95%ile 1-day) VaR of the asset can be expressed as

$$VaR_{0.05} = 1.645 \times \sigma \times V_t$$

- If returns are assumed to be normal, then the (99%ile 1-day) VaR of the asset can be expressed as

$$VaR_{0.01} = 2.326 \times \sigma \times V_t$$

- We will now turn our focus to obtaining the “elusive” σ . We will discuss few techniques that are currently popular
 - Standard deviation from historical data
 - RiskMetrics forecasting methodology
 - GARCH methodology

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Taking first step towards RiskMetrics

- We have essentially incorporated the three key assumptions
 - Assumptions 1 & 3: Conditional mean is zero and there is no autocorrelation in return series
 - Assumption 2: Volatility has been made variable; with an appropriate choice of model for volatility, the clustering effect can be captured

$$r_{t+1} = \sigma_{t+1} \varepsilon_{t+1} \quad \text{where} \quad \varepsilon_{t+1} \sim N(0,1)$$

- We essentially assume returns to be conditionally normally distributed
 - Why conditionally? The variance for time t+1 will “depend” on data till time t.
- We now have a time-varying volatility in our model.

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Volatility forecast: RiskMetrics Approach

- Let us quickly review the standard variance measure. We specify the below:

$$r_{t+1} = \sigma_{t+1} \varepsilon_{t+1} \quad \text{where} \quad \varepsilon_{t+1} \sim N(0,1)$$

- While using a rolling window (say m-day) to forecast volatility, we adopt

$$\sigma_{t+1}^2 = \frac{1}{N} \sum_{i=0}^{m-1} (R_{t-i} - \bar{R})^2$$

- To keep notations simple (and to be in-line with our assumptions), I am going to drop the mean term

$$\sigma_{t+1}^2 = \frac{1}{N} \sum_{i=0}^{m-1} (R_{t-i})^2$$

- What we are implicitly doing is assign equal weights to all terms, irrespective of whether they happened yesterday or 252 days before!
 - As should be evident, this shows up as the lag in the volatility forecast.

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Volatility forecast: RiskMetrics Approach

- What if instead of assigning equal weights to all terms, we assign greater weights to recent shocks and lower weights to past shocks?
 - That's precisely what JP Morgan's RiskMetrics methodology does
- Instead of assigning equal weights to all terms like standard variance

$$\sigma_{t+1}^2 = \frac{1}{N} \sum_{i=0}^{m-1} (R_{t-i})^2$$

- They assign exponentially decreasing weights as below

$$\sigma_{t+1}^2 = (1 - \lambda) \sum_{i=0}^{\infty} \lambda^i R_{t-i}^2$$

- To obtain a greater intuitive interpretation to the above, let us rearrange the terms a bit.

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Volatility forecast: RiskMetrics Approach

- RiskMetrics/EWMA model assigns exponentially decreasing weights as

$$\sigma_{t+1}^2 = (1 - \lambda) \sum_{i=0}^{\infty} \lambda^i R_{t-i}^2$$

- Let us separate the most recent term (i.e., $i=0$) from the summation

$$\sigma_{t+1}^2 = (1 - \lambda) \sum_{i=1}^{\infty} \lambda^i R_{t-i}^2 + (1 - \lambda) R_t^2$$

- Let us play around with these terms a bit to get a better understanding of what is that we are doing here.

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Volatility forecast: RiskMetrics Approach

- The RiskMetrics model can be written as:

$$\sigma_{t+1}^2 = (1 - \lambda) \sum_{i=1}^{\infty} \lambda^i R_{t-i}^2 + (1 - \lambda) R_t^2$$

$$\sigma_{t+1}^2 = \lambda(1 - \lambda) \sum_{i=0}^{\infty} \lambda^i R_{t-i-1}^2 + (1 - \lambda) R_t^2$$

- Now, how will variance forecast for day t look alike?

$$\sigma_t^2 = (1 - \lambda) \sum_{i=0}^{\infty} \lambda^i R_{t-i-1}^2$$

- Combining the above, we can rewrite variance forecast for t+1 as

$$\sigma_{t+1}^2 = \lambda \sigma_t^2 + (1 - \lambda) R_t^2$$